

3

Exercício 3.1

$f(x_-, y_-) = x^2 y;$

- a) $\partial_x f(0, 0) = 0$
- b) $\partial_x f(x_0, y_0) = 2 x_0 y_0$
- c) $\partial_y f(1, 2) = 1$
- d) $\partial_y f(x_0, y_0) = x_0^2$

Exercício 3.2

a)

$f(x_-, y_-) = 3 x^2 + 2 y^2;$

$\partial_x f = 6 x$
 $\partial_y f = 4 y$

b)

$f(x_-, y_-) = \sin(x^2 - 3 x y);$

$\partial_x f = (2 x - 3 y) \cos(x^2 - 3 x y)$
 $\partial_y f = -3 x \cos(x^2 - 3 x y)$

c)

$f(x_-, y_-) = x^2 y^2 \exp(2 x y);$

$\partial_x f = 2 e^{2 x y} x y^2 + 2 e^{2 x y} x^2 y^3$
 $\partial_y f = 2 e^{2 x y} x^2 y + 2 e^{2 x y} x^3 y^2$

d)

$f(x_-, y_-) = \exp(x) \log(x y);$

$\partial_x f = \frac{e^x}{x} + e^x \log(x y)$
 $\partial_y f = \frac{e^x}{y}$

e)

$f(x_-, y_-) = \exp(\sin(x \sqrt{y}));$

$\partial_x f = e^{\sin(x \sqrt{y})} \sqrt{y} \cos(x \sqrt{y})$
 $\partial_y f = \frac{e^{\sin(x \sqrt{y})} x \cos(x \sqrt{y})}{2 \sqrt{y}}$

f)

$f(x_-, y_-) = \frac{x^2 + y^2}{x^2 - y^2};$

$\partial_x f = -\frac{4 x y^2}{(x^2 - y^2)^2}$
 $\partial_y f = \frac{4 x^2 y}{(x^2 - y^2)^2}$

g)

$f(x_-, y_-) = x \cos(x) \cos(y);$

$\partial_x f = \cos(y) (\cos(x) - x \sin(x))$
 $\partial_y f = -x \cos(x) \sin(y)$

h)

$f(x_-, y_-) = \text{ArcTan}(x^2 y^3);$

$\partial_x f = \frac{2 x y^3}{1 + x^4 y^6}$
 $\partial_y f = \frac{3 x^2 y^2}{1 + x^4 y^6}$

i)

$f(x_-, y_-) = x + x y^2 + \log(\sin(x^2 + y));$

$\partial_x f = 1 + y^2 + 2 x \cot(x^2 + y)$
 $\partial_y f = 2 x y + \cot(x^2 + y)$

j)

$f(x_-, y_-, z_-) = z \exp(x^2 + y^2);$

$\partial_x f = 2 e^{x^2 + y^2} x z$
 $\partial_y f = 2 e^{x^2 + y^2} y z$

$\partial_z f = e^{x^2 \cdot y^2}$

k)

$f[x_, y_, z_] = \text{Log}[\text{Exp}[x] + z^y];$

$$\partial_x f = \frac{e^x}{e^x + z^y}$$
$$\partial_y f = \frac{z^y \text{Log}[z]}{e^x + z^y}$$
$$\partial_z f = \frac{y z^{-1+y}}{e^x + z^y}$$

l)

$f[x_, y_, z_] = \frac{x y^3 + \text{Exp}[z]}{x^3 y - \text{Exp}[z]};$

$$\partial_x f = -\frac{y \left(2 x^3 y^3 + e^z \left(3 x^2 + y^2\right)\right)}{\left(e^z - x^3 y\right)^2}$$
$$\partial_y f = \frac{2 x^4 y^3 - e^z x \left(x^2 + 3 y^2\right)}{\left(e^z - x^3 y\right)^2}$$
$$\partial_z f = \frac{e^z x y \left(x^2 + y^2\right)}{\left(e^z - x^3 y\right)^2}$$

Exercício 3.3

a)

$\partial_x f(0, 0) = \text{Não existe}$
 $\partial_y f(0, 0) = \text{Não existe}$

b)

$\partial_x f(0, 0) = 0$
 $\partial_y f(0, 0) = 0$

Exercício 3.4

a)

$f[x_, y_] = \text{Exp}[x y];$

$x \partial_x f = e^{x y} x y$
 $y \partial_y f = e^{x y} x y$

b)

$f[x_, y_] = \text{Log}[x^2 + y^2 + x y];$

$$x \partial_x f = \frac{x \left(2 x + y\right)}{x^2 + x y + y^2}$$
$$y \partial_y f = \frac{y \left(x + 2 y\right)}{x^2 + x y + y^2}$$

$x \partial_x f + y \partial_y f = 2$

c)

$f[x_, y_, z_] = x + \frac{x - y}{y - z};$

$$\partial_x f = 1 + \frac{1}{y - z}$$
$$\partial_y f = \frac{-x + z}{\left(y - z\right)^2}$$
$$\partial_z f = \frac{x - y}{\left(y - z\right)^2}$$

$\partial_x f + \partial_y f + \partial_z f = 1$

Exercício 3.5

$\text{Ddirecao}[f_, P_, u_] := \text{Limit}\left[\frac{f \text{@@} (P + h u) - f \text{@@} P}{h}, h \rightarrow 0\right];$

a)

$f[x_, y_] = x^2 y + x; P = \{1, 0\}; u = \{1, 1\};$

$\text{Ddirecao}[f, P, \text{Normalize}[u]]$

$\sqrt{2}$

b)

$f[x_, y_] = x^2 \sin[2 y]; P = \left\{1, \frac{\pi}{2}\right\}; u = \{3, -4\};$

$\text{Ddirecao}[f, P, \text{Normalize}[u]]$

$-\frac{8}{5}$

c)

$f[x_, y_, z_] = x^2 + y^2 + z^2; P = \{1, 2, 3\}; u = \{1, 1, 1\};$

$\text{Ddirecao}[f, P, \text{Normalize}[u]]$

$4 \sqrt{3}$

Exercício 3.6

- a)

Grad[x Exp[-x + y], {x, y}]

$\left\{e^{-x+y}-e^{-x+y}x, e^{-x+y}x\right\}$
- b)

Grad[r Sin[θ], {r, θ}]

$\left\{\text{Sin}\left[\theta\right], r\text{Cos}\left[\theta\right]\right\}$
- c)

Grad[π r^2 h, {r, h}]

$\left\{2h\pi r, \pi r^2\right\}$
- d)

Grad[x Exp[-x^2 - y^2 - z^2], {x, y, z}]

$\left\{e^{-x^2-y^2-z^2}-2e^{-x^2-y^2-z^2}x^2, -2e^{-x^2-y^2-z^2}xy, -2e^{-x^2-y^2-z^2}xz\right\}$
- e)

Grad[$\frac{xyz}{x^2+y^2+z^2+1}$, {x, y, z}] // Simplify

$\left\{\frac{yz\left(1-x^2+y^2+z^2\right)}{\left(1+x^2+y^2+z^2\right)^2}, \frac{xz\left(1+x^2-y^2+z^2\right)}{\left(1+x^2+y^2+z^2\right)^2}, \frac{xy\left(1+x^2+y^2-z^2\right)}{\left(1+x^2+y^2+z^2\right)^2}\right\}$
- f)

Grad[z^2 Exp[x] Cos[y], {x, y, z}] // Simplify

$\left\{e^x z^2\text{Cos}\left[y\right], -e^x z^2\text{Sin}\left[y\right], 2e^x z\text{Cos}\left[y\right]\right\}$

Exercício 3.7

$f[x_, y_] = x^2 + y^3;$

Equação do plano tangente: z = -11+6 x+3 y

Exercício 3.8

$f[x_, y_] = x^2 + y^2;$

Equação do plano tangente a f em (0,0): z = 0

$g[x_, y_] = -x^2 - y^2 + x y^3;$

Equação do plano tangente a g em (0,0): z = 0

Exercício 3.9

$f[x_, y_] = \text{Exp}[x^2 - y^2];$

a)

Equação do plano tangente a f em (1,1): z = 1+2 x-2 y

Ponto de interseção do plano tangente com o eixo dos zz: {{x→0, y→0, z→1}}

b)

Cota do ponto do plano tangente correspondente a x=y=1: z=1.24

$f[1, 1]$

1

Exercício 3.10

Ver diapositivos

Exercício 3.11

Ver diapositivos

Exercício 3.12

- a)

$f[x_, y_] = \frac{x^2 y}{x^2 + y^2};$

$f[0, 0] = 0;$

$Df((0,0);(u1,u2))=\frac{u1^2 u2}{u1^2 + u2^2}$

Se (x,y)≠(0,0) então $Df((x,y);(u1,u2))=\frac{x\left(2u1y^3+u2\left(x^3-x y^3\right)\right)}{\left(x^2+y^2\right)^2}$
- b)

f não é diferenciável na origem

Exercício 3.13

a)

$$f(x,y) = (x^2 + y^2) \sin\left(\frac{1}{\sqrt{x^2 + y^2}}\right),$$

$$f(0,0) = 0,$$

$$\partial_x f(0,0) = 0$$

$$\partial_y f(0,0) = 0$$

b)

$$\partial_x f(x,y) = -\frac{x \cos\left(\frac{1}{\sqrt{x^2 + y^2}}\right)}{\sqrt{x^2 + y^2}} + 2x \sin\left(\frac{1}{\sqrt{x^2 + y^2}}\right)$$

$$\partial_y f(x,y) = -\frac{y \cos\left(\frac{1}{\sqrt{x^2 + y^2}}\right)}{\sqrt{x^2 + y^2}} + 2y \sin\left(\frac{1}{\sqrt{x^2 + y^2}}\right)$$